



## Koneru Lakshmaiah Education Foundation

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### STUDENT CASE STUDY EXECUTION REPORT

|                |                          |
|----------------|--------------------------|
| Department     | FED                      |
| Subject        | DSA-2                    |
| Academic Year  | I-III Semester (2025-26) |
| Faculty Name   | Dr.Swetha Nagasri        |
| Student Name   | V Sameer Reddy           |
| Roll Number    | 2520030276               |
| Branch         | CSE                      |
| Course Outcome | CO-4                     |

#### Case Study Topic: BMTC Bus Route Shortest Path using Bellman-Ford Algorithm

##### Work Done Summary:

In this case study, we developed a Java-based shortest path system for the Bangalore Metropolitan Transport Corporation (BMTC) route network. The system calculates the shortest travel time from Majestic (MJC) to all other bus hubs during peak hours. Since one route contains a negative edge weight (Whitefield to Marathahalli = -3 minutes), the Bellman-Ford algorithm was used instead of Dijkstra's algorithm.

Using Java, we implemented graph representation with directed weighted edges and applied the Bellman-Ford algorithm for shortest path computation. This algorithm supports negative edge weights and can also detect negative cycles.

The Bellman-Ford approach ensures accurate shortest path calculations even in complex real-world traffic systems where certain routes may reduce travel time unexpectedly.

##### Implementation Details:

1. Created an **Edge class** in Java to store source, destination, and weight.
2. Represented BMTC bus hubs as graph vertices.
3. Added 11 directed edges with travel times and surge penalties.
4. Implemented Bellman-Ford algorithm with **V-1 iterations** for relaxation.

5.

6.

7. Updated shortest distances after each iteration.
8. Added negative cycle detection using one extra relaxation pass.
9. Displayed shortest distance from source hub (MJC) to all other hubs.

### Result: Output Screenshots:

```
ExceptionMessages -cp C:\Users\malli\AppData\Roaming\Code\User\ws\Case Study 4_90b96818\bin' 'BMTCBellmanFord'
Shortest distance from MJC:
MJC -> 0
KEM -> 8
JAY -> 5
KOR -> 9
WHF -> 15
HBR -> 15
MRT -> 12
PS C:\Users\malli\OneDrive\Documents\KLH\DSA\2\Case Study 4>
```

Time Complexity:

- Bellman - Ford Algorithm:  $O(V \times E)$
- Here:  $V=7$  ;  $E = 11$
- Total operations =  $7 * 11 = 77$

SLA Analysis:

- Total Bus Hubs = **7**
- Total Routes = **11**
- Source Hub = **MJC**
- Negative Edge Present = **1**
- Negative Cycle = **No**
- Total Relaxation Iterations = **6**

**GitHub Link:** <https://github.com/sameerreddyvemireddy/DSA-Case-STUDIES>

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|-------------------|-------------------|
| Student Signature | Faculty Signature |
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